



BATCH : B146-149 **CS & DevOps**
LESSON : **Network**
DATE : 08.07.2023
SUBJECT : **Recap & CPT Practice**

ZOOM GİRİŞLERİNİZİ LÜTFEN **LMS** SİSTEMİ ÜZERİNDEN YAPINIZ





Previous Session

Addressing
Subnetting
CIDR



NETWORK Day 7

Contents

- Recap
- CPT Practice



What is a Network?



- ❓ A computer network is a group of computers that use a set of common communication protocols over digital inter-connections for the purpose of sharing resources located on or provided by the network nodes.



Types of Networks

Geographical

- NANO
- BAN
- PAN
- LAN
- CAN
- MAN
- WAN

Network Architecture

- Client–Server
- P2P

Topological

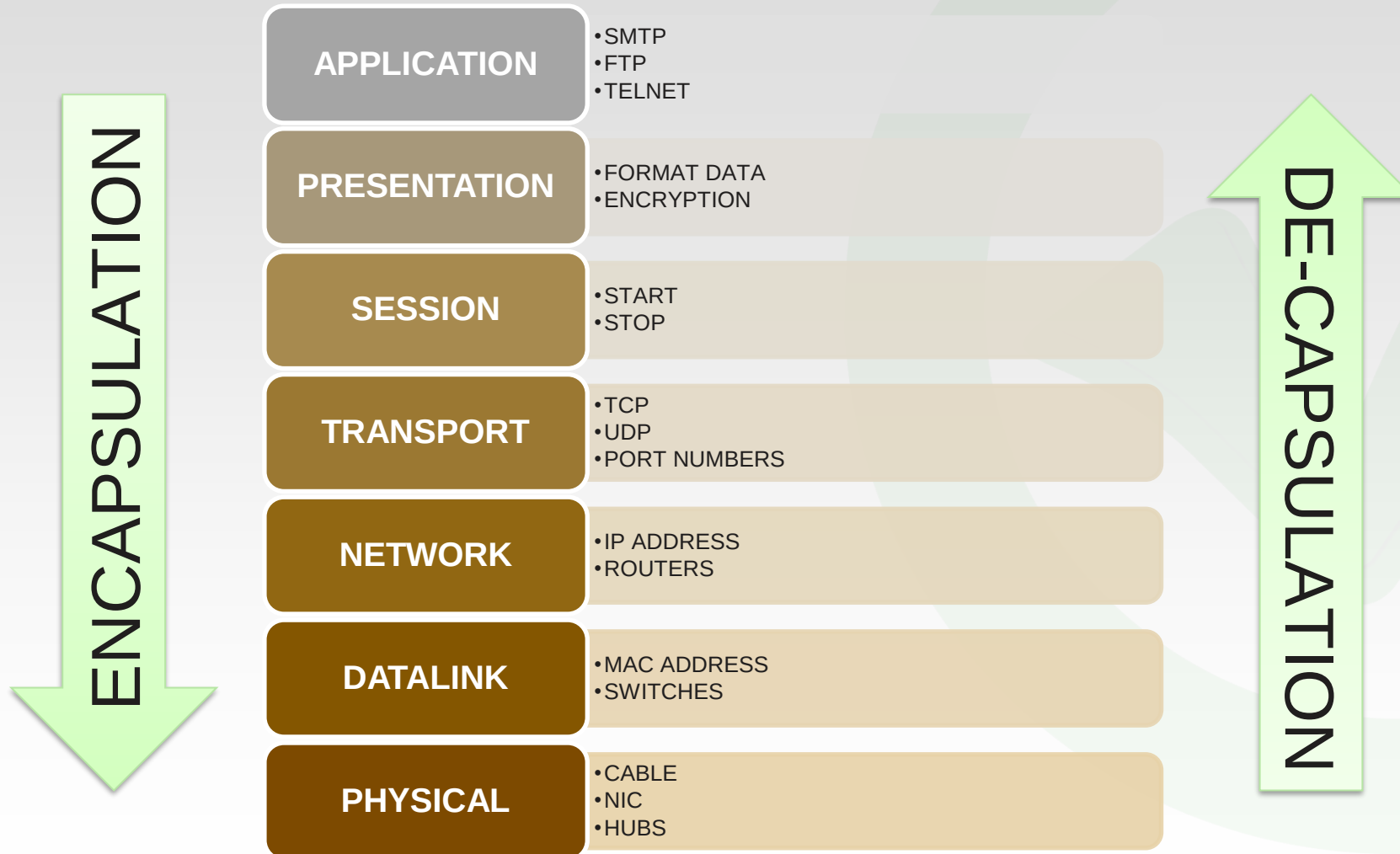
- Ring
- Star
- Mesh
- Bus
- Line

Transferring Mediums

- Cable
- Wireless
 - RF
 - Laser
 - Microwave



OSI Reference Model





Transmission Media

❖ Cable

- Coaxial
- Twisted pair
- Fiber

❖ Wireless

- Laser (WLAN)
- Infrared (Bluetooth)
- Radio Frequency (WLAN, Satellite)



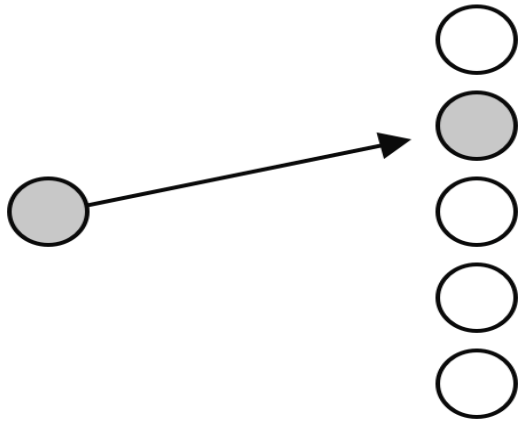
Network Devices

- ❖ Network Interface Card (NIC)
- ❖ Hub
- ❖ Bridge
- ❖ Switch
- ❖ Router
- ❖ Firewall
- ❖ DNS
- ❖ DHCP
- ❖ Other specialized devices

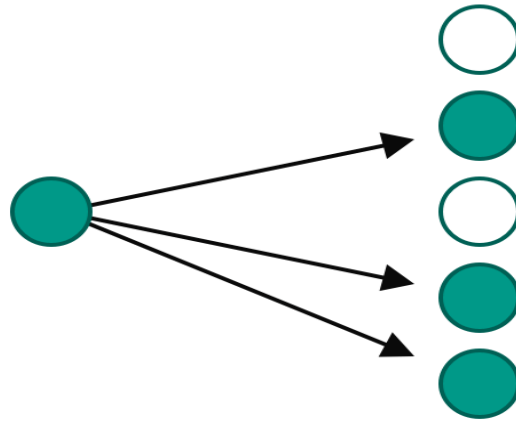


Data Transmission

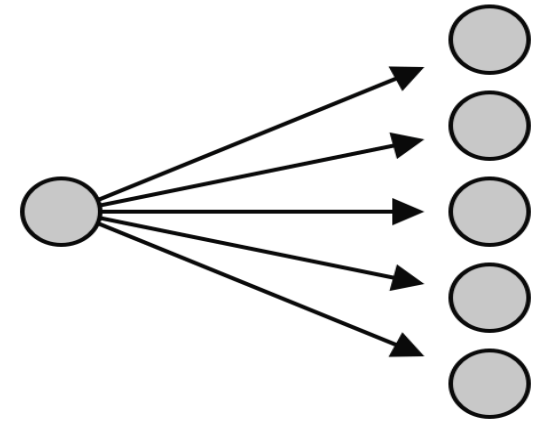
Multicasting



Unicast



Multicast

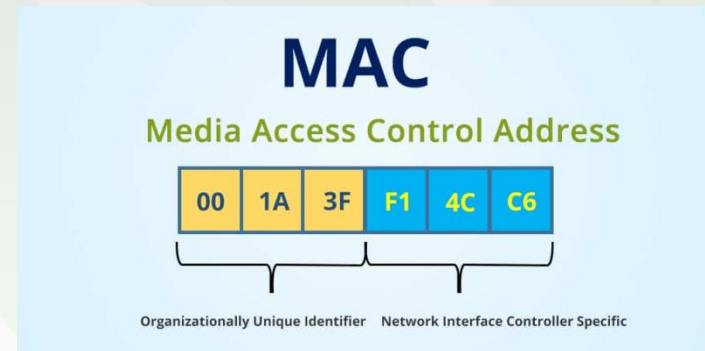


Broadcast



MAC Address

- MAC adres dağıtımları **IEEE** tarafından yönetilir.
- MAC**, 48 **bit**'lik bir **adres** olduğundan dolayı $2^{48} = 281,474,976,710,656$ değişik ağ kartını tanımlamak için kullanılabilir. **MAC adresi** (Fiziksel **adres**, Donanım **adres**i), ağ donanımının tanımlanmasını sağlar. **MAC adresi**, bilgisayarın ethernet kartına üretici tarafından kodlanan bir bilgidir.
- Üreticiler **MAC** adres aralıklarını satın alırlar.
 - Aynı ağ içerisinde birbirine fiziksel olarak bağlı birimler arasında çerçeve transferinde kullanılır.



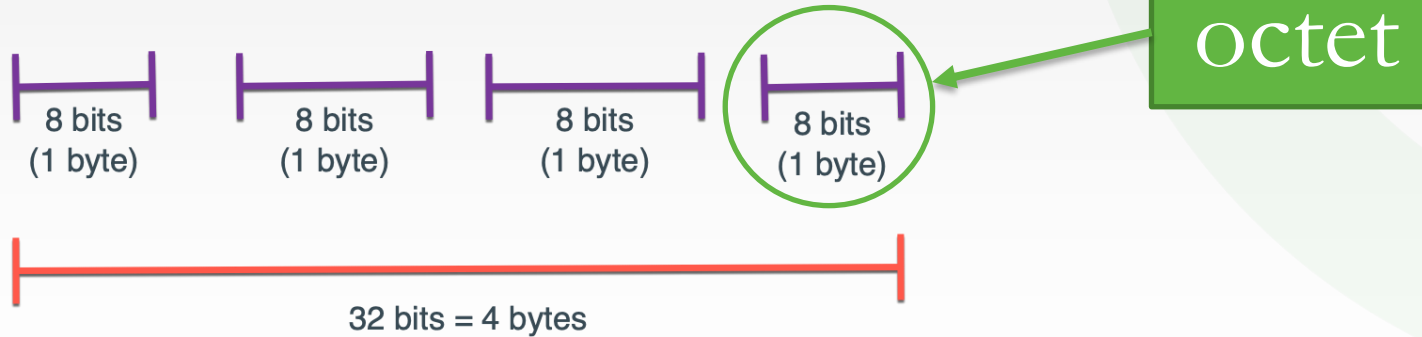


IP Address

❓ IP adres:

- ❓ Network Layer Address
- ❓ Data Packet – Segment – Datagram
- ❓ **32-bit** → **17.172.224.47 (IPv.4)**

17.172.224.47





SubnetMask

IP address **192.168.0.96** and Mask **255.255.255.0**

Binary IP:

11000000.10101000.00000000.01100000

Binary Mask:

11111111.11111111.11111111.00000000

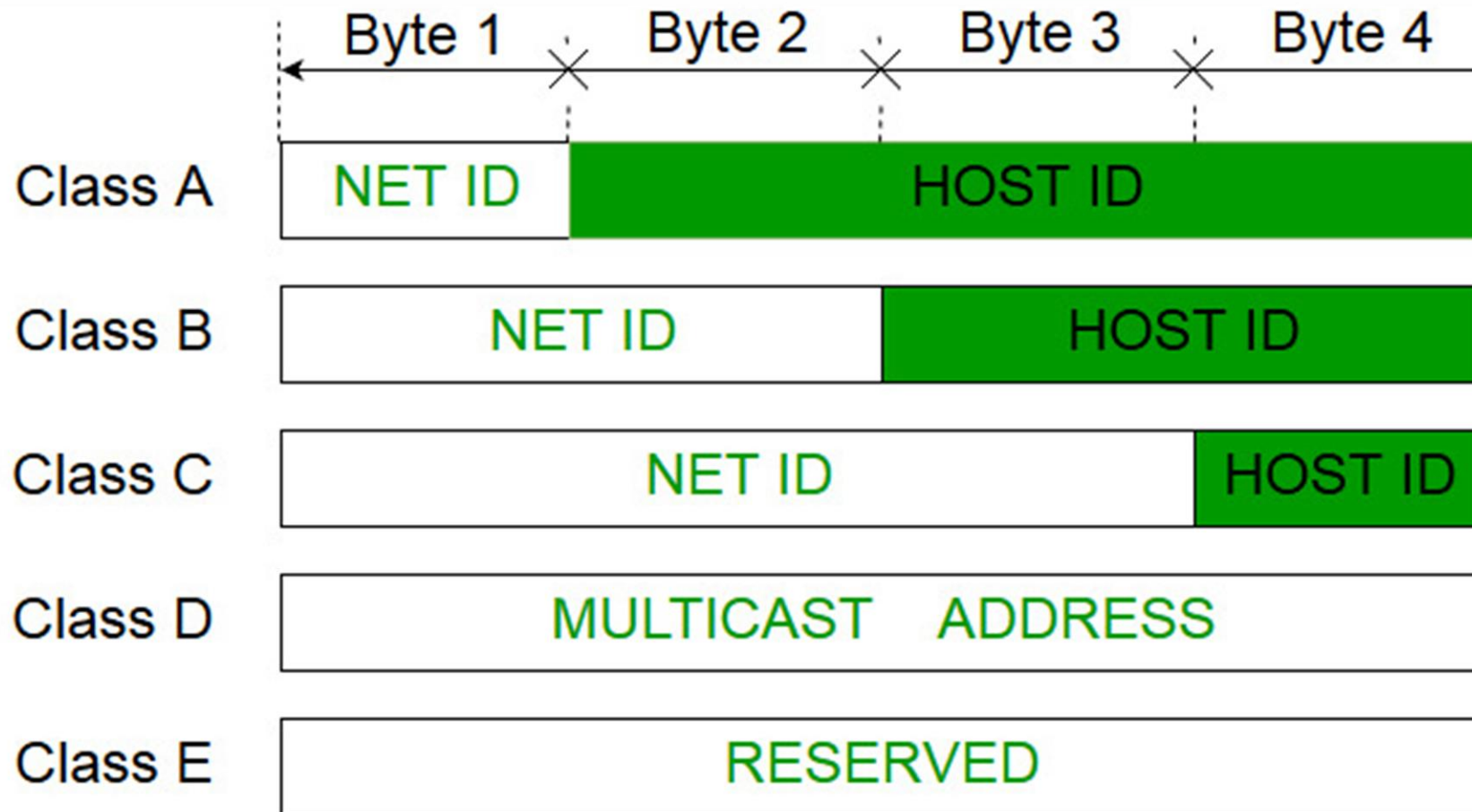
Network
Field

Host
Field

Figure 3.1 IP address and mask in binary, showing network and host fields.



IP Classification





IP Classification

Class	Leading bits	Size of <i>network number</i> bit field	Size of <i>rest bit</i> field	Number of networks	Addresses per network	Total addresses in class	Start address	End address
Class A	0	8	24	128 (2^7)	16,777,216 (2^{24})	2,147,483,648 (2^{31})	0.0.0.0	127.255.255.255
Class B	10	16	16	16,384 (2^{14})	65,536 (2^{16})	1,073,741,824 (2^{30})	128.0.0.0	191.255.255.255
Class C	110	24	8	2,097,152 (2^{21})	256 (2^8)	536,870,912 (2^{29})	192.0.0.0	223.255.255.255
Class D (multicast)	1110	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	224.0.0.0	239.255.255.255
Class E (reserved)	1111	not defined	not defined	not defined	not defined	268,435,456 (2^{28})	240.0.0.0	255.255.255.255

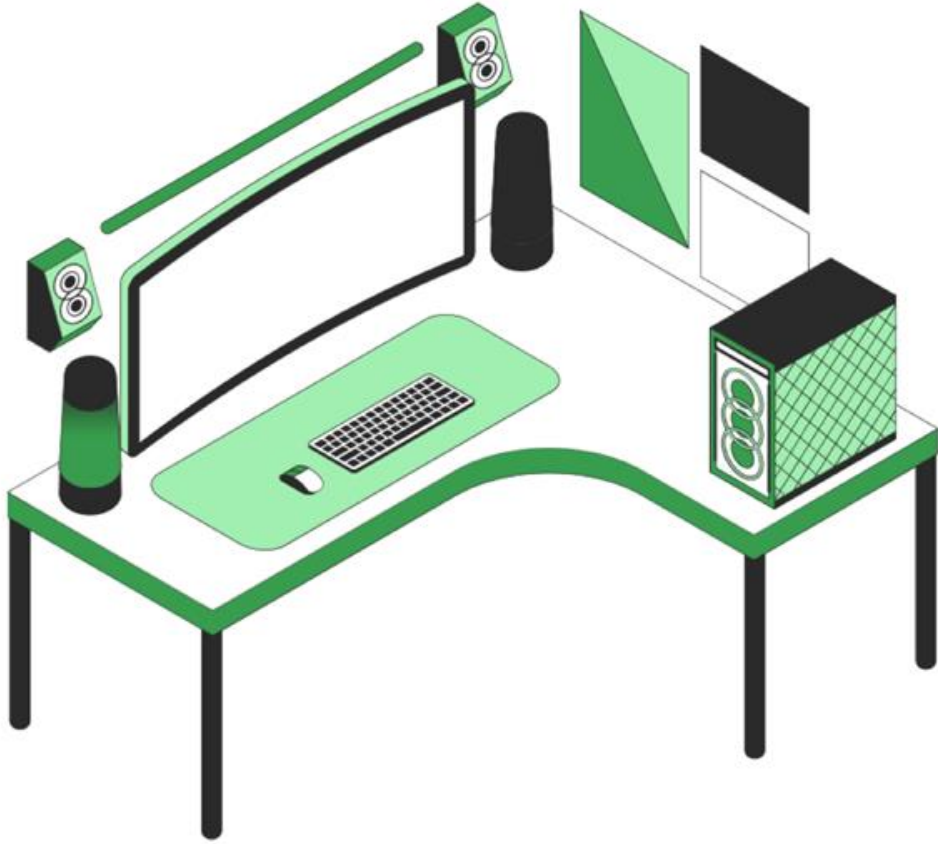


Private Networks

Public and Private IP Addresses

- No two machines that connect to a public network can have the same IP address because public IP addresses are global and standardized.
- However, private networks that are not connected to the Internet may use any host addresses, as long as each host within the private network is unique.
- RFC 1918 sets aside three blocks of IP addresses for private, internal use.
- Connecting a network using private addresses to the Internet requires translation of the private addresses to public addresses using Network Address Translation (NAT).

Class	RFC 1918 internal address range
A	10.0.0.0 to 10.255.255.255
B	172.16.0.0 to 172.31.255.255
C	192.168.0.0 to 192.168.255.255



Do you
have any
questions?

Send it to us! We hope you learned
something new.